

MPLADS IntelliTrack

Complete Solution Blueprint & System Specification

AI-Powered Monitoring and Decision-Support Platform for MPLADS Governance

1. Core Solution & Fundamental Principle

MPLADS IntelliTrack is an AI-powered monitoring and decision-support platform designed to establish a verifiable digital evidence trail for every Member of Parliament Local Area Development Scheme (MPLADS) work. The system continuously tracks project, financial, execution, and compliance metrics to detect anomalies, predict cost overruns and delays, and deliver explainable alerts to authorized governance bodies.

Fundamental Governance Principle:

AI detects → Evidence supports → Authority verifies → Governance acts

The system does not replace administrative procedures, sanction funds automatically, or declare fraud. Human verification remains mandatory for all escalations.

2. Complete End-to-End Workflow

1. Recommendation & Sanction: MP recommends work → District Authority scrutinizes → District Authority sanctions/approves → Implementing Agency assigned.

2. Execution & Evidence: Tender / Procurement → Contractor selected → Project execution → Progress, expenditure, bills, and geofenced photographic evidence submitted.

3. AI Intelligence & Verification: Data validation → AI Analytics Engine (Anomaly + Prediction + Compliance) → Risk Score → Explainable Alert → Responsible Authority → Field Verification → Corrective Action / Closure → Audit Trail.

3. Core System Modules (10 Pillars)

- **Module 1 — Project Management:** Registration, MP recommendation, sanction info, unique project IDs, geospatial location, categories, estimates, and milestones.
- **Module 2 — Digital Evidence:** In-app camera, GPS validation, timestamps, project geofencing, bills, progress photos, and inspection logs.
- **Module 3 — Financial Intelligence:** Expenditure monitoring, cost benchmarking, invoice verification, material-rate comparison, and duplicate invoice detection.
- **Module 4 — Execution Intelligence:** Milestone tracking, physical vs. financial progress comparison, delay detection, and project health scoring.
- **Module 5 — AI Anomaly Detection:** Automated scanning for cost, financial, duplicate, invoice, progress, and evidence anomalies.
- **Module 6 — Predictive Intelligence:** Advanced modeling for delay probability, cost-overruns, risk trajectories, and contractor pattern analysis.
- **Module 7 — Risk Engine:** Multi-factor risk scoring, severity classification, explainable AI rationales, and historical risk tracking.
- **Module 8 — Decision Support:** Smart alerting, automated recommendations, verification workflows, and corrective-action tracking.
- **Module 9 — Governance:** Role-Based Access Control (RBAC), jurisdiction enforcement, immutable audit trails, and data integrity safeguards.
- **Module 10 — Analytics & Dashboards:** Unified command centers for MoSPI, State, District, MP, Implementing Agencies, Contractors, and GIS maps.

4. Data Architecture Layers

To satisfy strict government audit standards, information is segregated into four distinct architectural layers:

Data Layer	Description	Examples
A. Source Data	Raw submitted and recorded data	Sanction amount, invoices, payments, GPS coordinates, photographs
B. Derived Data	Deterministic system calculations	Cost deviation %, expenditure %, delay days, geospatial distance
C. AI Inference	Model-generated statistical conclusions	Anomaly probability, delay score, duplicate similarity, risk score
D. Human Decision	Official administrative outcomes	Verified, corrective action initiated, escalated, closed, under investigation

5. Database Schema & Project ID Standard

Every related artifact in the lifecycle ties back to a standardized **Unique Project ID**:

Format: MPLADS- [STATE] - [DISTRICT] - [YEAR] - [SERIAL]

Example: MPLADS-MH-AMR-2026-00452

Relational hierarchy:

USER → Role / State / District → Jurisdiction

PROJECT → Recommendation → Sanction → Location → Milestones → Agency → Contractor

PROJECT → [EXPENDITURE | INVOICE | EVIDENCE | PROGRESS | INSPECTION | ANOMALY |

RISK_SCORE | ALERT | AUDIT_LOG]

6. AI & Machine Learning Methodology

7. Cost Anomaly Detection

Utilizes statistical benchmarking and Isolation Forest / XGBoost regression models. Compares project type, capacity, location, materials, and labor against historical comparable projects to compute deviation percentages.

Expected Cost: ₹43.2L | *Actual Estimate:* ₹50L

Deviation: +15.7% → **HIGH RISK**

8. Progress-Expenditure Mismatch

Calculates financial progress (*Expenditure / Sanction*) and contrasts it against physical milestone completion, flagging uncoordinated capital outflows.

Physical Progress: 42%

Financial Progress: 83%

Mismatch: 41% gap → **ALERT**

9. Delay Prediction

XGBoost / Random Forest classifier evaluating planned completion dates, progress velocity, milestone lag, and expenditure burn rates to forecast timeline slippages.

Delay Probability: 78%

Key Drivers: Milestone lag, low velocity.

10. Duplicate Project Detection

Combines NLP sentence embeddings for project descriptions with PostGIS geospatial proximity matching and structured parameter checks to prevent overlapping fund allocations.

NLP Similarity: 89% | *Geo Proximity:* 97%

Result: Potential duplicate work flagged.

11. Invoice & Evidence Intelligence

Invoices pass through an OCR and document parsing pipeline to extract itemized Bill of Quantities (BOQ), comparing unit rates against regional benchmark price bands. Photographic evidence is automatically validated for GPS coordinate proximity, temporal capture windows, hash uniqueness (anti-duplication), and visual consistency across project stages.

12. Risk Scoring & Explainability

The Risk Engine aggregates normalized scores (0–100) across financial, cost, progress, delay, evidence, compliance, and duplication dimensions into an overall risk index.

Risk Tier	Score Range	Governance Action
Normal	0 – 30	Standard automated tracking. No action required.
Watch	31 – 50	Minor deviation noted. Monitored in routine cycle.
Review	51 – 75	Requires closer administrative examination.
High Risk	76 – 100	Prioritized human verification and field inspection.

Explainable AI Output Example (Project ID: MPLADS-MH-AMR-2026-00452):

Overall Risk: 76 / 100 — HIGH

- Cost is 18% above comparable regional projects.
- Financial progress (78%) significantly outpaces physical progress (46%).
- Current velocity indicates a 72% probability of delay.
- **Recommendation:** Verify BOQ specifications, expenditure documents, and current site status.

13. Dashboards, GIS, & Contractor Intelligence

Role-based workspaces provide tailored analytical depth:

- **MoSPI / State / District:** National command centers, regional comparative metrics, utilization heatmaps, and systemic anomaly tracking via interactive GIS maps.
- **MP / Implementing Agency:** Constituency progress tracking, fund utilization, pending verifications, and milestone sign-offs.
- **Contractor Workspace:** Streamlined portal for geocamped progress uploads, milestone reporting, and digital invoice submissions.
- **Contractor Performance Profiles:** Tracks historical on-time completion rates, average delays, cost deviations, and anomaly frequencies as risk indicators rather than blacklists.

14. Security, Fraud Prevention, & Edge Cases

Security Architecture: Enforces secure authentication, RBAC jurisdiction control, data encryption in transit and at rest, and an immutable audit trail recording every administrative modification.

Fraud Prevention Philosophy: Operates on three distinct tiers: *Level 1 (Anomaly)* → *Level 2 (Risk)* → *Level 3 (Human Verification)*. Only authorized government officers determine final administrative actions.

Edge Cases Handled: Offline GPS caching with later sync, flagged evidence for coordinate mismatches (avoiding hard rejections), project-type aware expenditure curves (accounting for front-loaded costs), and false-positive feedback loops for continuous model retraining.

15. SIH Prototype Scope & Tech Stack

Suggested Technology Stack

- **Frontend:** React.js / Next.js (Dashboards & Portals)
- **Backend:** Java Spring Boot (Enterprise REST APIs)
- **AI Services:** Python FastAPI + Scikit-learn, XGBoost, OpenCV, Sentence Transformers
- **Database & GIS:** PostgreSQL with PostGIS extension

Vertical Slice Prototype Flow

1. Register MPLADS project & assign agency.
2. Contractor submits progress, photo, GPS, invoice.
3. System validates evidence & runs AI models.
4. Demonstrate 5 core anomalies (Cost, Progress, Delay, Duplicate, Invoice).
5. Generate Risk Score & Explainable Alert.
6. Route alert to District Authority for verification & audit logging.

MPLADS IntelliTrack Solution Blueprint • Prepared for Smart India Hackathon (SIH)